Goals

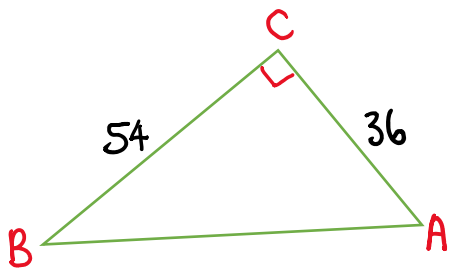
- Understand and apply these definitions: angles in standard position (from 0 to 360 degrees), reference angle (from 0 to 90 degrees), coterminal angles (outside the range of 0 to 360 degrees), terminal arm, 2 special triangles
- Determine the possible triangles that can be draw: 0, 1, or 2 with ambiguous sine law cases
- Use sine law and cosine law to solve non-right triangles with or without word problems
- Determine exact values of trigonometric ratios
- Find angles from trigonometric ratios in various quadrants given specific domains
- Solve trigonometric equations

Special Notes:

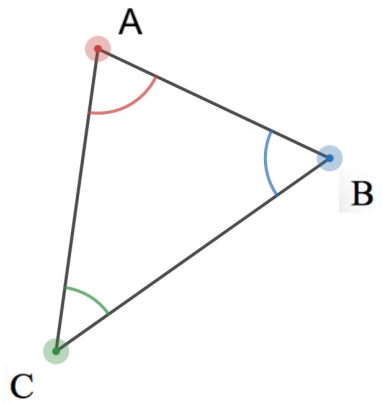
1) Check out Content -> Chapter 6: Trigonometry -> Supplemented Materials: Summary of Key Ideas on Trigonometry for some sample questions and answers to this assignment.

Please answer all questions on this sheet and remember to show all your work!
For all the questions, students should try to keep as many decimal places as possible for the steps and round only until the very end. Unless specified, all the answers should be <u>rounded to two decimal places</u>.

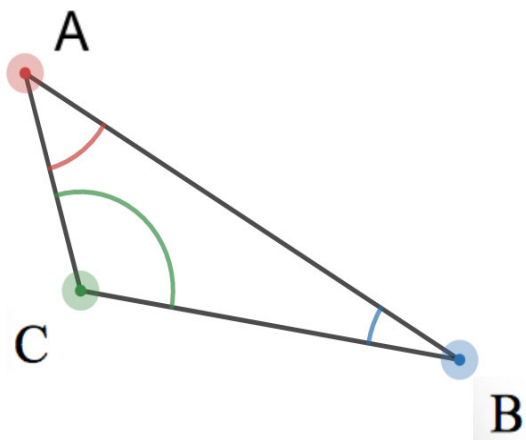
1. Solve the following triangle (find every missing side and angle):



2. Solve the following triangles using the Sine Law and the Cosine Law.
   a. ΔABC, where $\angle B = 50°, \overline{AC} = 45$, and $\overline{AB} = 38.9$



b. ΔABC, $\overline{AB} = 27.4, \overline{BC} = 19.8$, and $\overline{AC} = 11.4$

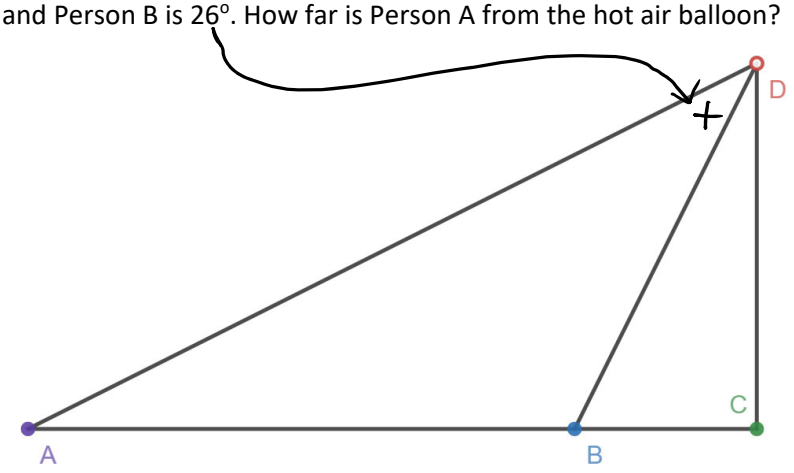


3. For each, **determine the number of unique triangles** that can be made from the following information.

   a. For $\Delta DEF, \angle F = 55°, \overline{DE} = 66,$ and $\overline{DF} = 78.$

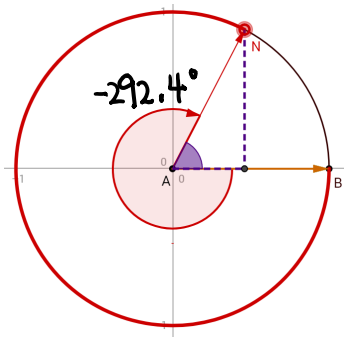
   b. For $\Delta LMN, \angle L = 41°, \overline{LM} = 7.2cm,$ and $\overline{MN} = 2.9cm.$

4. Three people are watching a hot air balloon travel over their town. At a certain point in time, one person stands directly below the balloon, and the others look at it at certain angles. In the following image, A, B, and C are people, and D is the balloon. Person C is 392m directly below the balloon, Person B is 225m away from Person C, and the angle between Person A, the balloon, and Person B is 26°. How far is Person A from the hot air balloon?

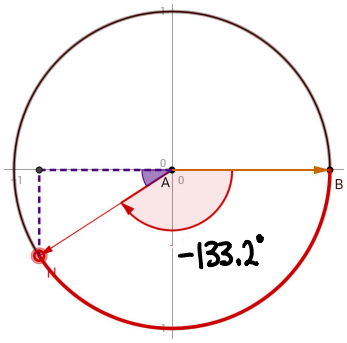


5. For each of the following terminal arms, determine:
   a) the angle in standard position ($0 < \theta < 360°$), **b)** quadrant (I, II, III or IV),
   c) reference angle ($0 < \theta < 90°$), and **d)** the coterminal angle (outside of $0 < \theta < 360°$)

   a.



   b.



6. Explain with words and math symbols how you can use special triangle or unit circle to solve for the exact value of $\sin 300°$ in standard position. Find the exact value of $\sin 300°$.

7. Suppose there is a coordinate $(-1, \sqrt{5})$ at the end of a terminal arm and θ represents the angle in standard position. Determine the **exact values** of $\sin\theta$, $\cos\theta$, and $\tan\theta$.

8. Solve the following, where $0° < \theta < 360°$.

   a. If $\tan\theta = \frac{5}{4}$ and $\sin\theta < 0$, find θ

   b. If $\frac{8}{3}\cos\theta + 7 = 5$, find θ.

Self Assessment

How well do you think you did on this assignment?

Where do you think your overall understanding of this chapter is with the following scale? (Circle one)

Emerging	Developing	Proficient	Extending
Student demonstrates a limited understanding of the concepts and competencies relevant to the expected learning.	Student demonstrates a basic understanding of the concepts and competencies relevant to the expected learning.	Student demonstrates a strong understanding of the concepts and competencies relevant to the expected learning.	Student demonstrates an extensive understanding of concepts and competencies relevant to the expected learning.

Is there an area or areas that you think you need more practice on? You can click all that apply using the chart below and/or add your own point(s).

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- ☐ Determine the possible triangles that can be draw: 0, 1, or 2 with ambiguous sine law cases
- ☐ Use sine law and cosine law to solve non-right triangles with or without word problems
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- ☐ Find angles from trigonometric ratios in various quadrants given specific domains
- ☐ Solve trigonometric equations
- ☐ Additional point #1:
- ☐ Additional point #2:
- ☐ Additional point #3:

Please combine all pages into 1 single PDF file for assignment submission. Thank you!

~The End~